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Capital Investment Decision-Making

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Capital Investment Decision-Making

Operational Excellence, Glencore Copper
30 May 2022, Version 1.0

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Supported by the AM Network Pty Ltd and Work Management Solutions Pty Ltd

Preface

The Asset Management Framework, Policy and Manuals are intended to provide leading Asset Management practices when applied and will be continually improved for the benefit of Glencore Copper.

The real value of sharing the knowledge gained from the use of these documents will come from continued discussion of the issues and rationale behind each document rather than by reading and interpreting them as individuals.

Readers are encouraged to consider these against our values around Safety, Responsibility, Simplicity, Integrity, Openness and Entrepreneurialism and how they reflect our Purpose – to responsibly source commodities for advancing everyday life. This document lays out the minimum standard for Glencore Copper and compliance to this manual will demonstrate a maturity value equal to Planned/Deployed (3) on the Glencore Copper Asset Management Maturity Scale.

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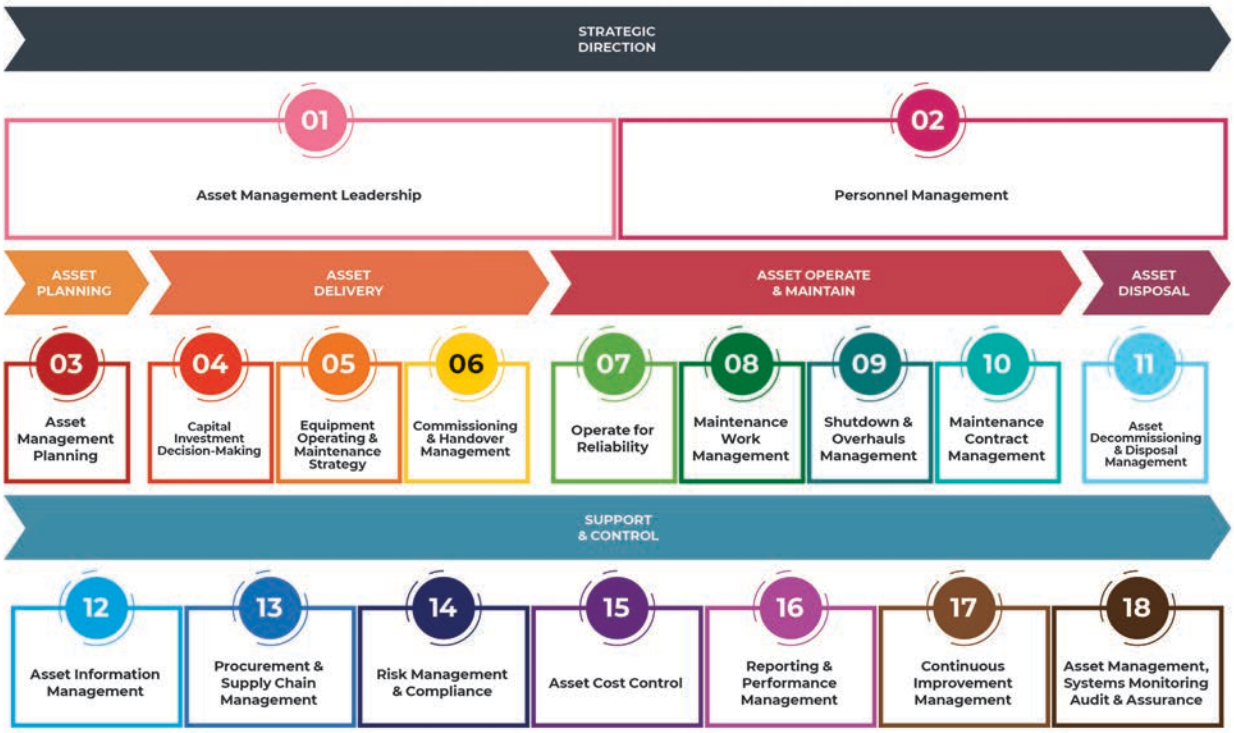
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1. Introduction

The Asset Management Policy and Manuals define how Glencore Copper manages the whole of life of an asset to ensure optimal value is extracted by ensuring assets are capable of meeting business and stakeholder objectives.

Figure 1-1 describes the high-level Asset Management Framework, which enables Glencore Copper to deliver its Asset Management purpose and vision.

FIGURE 1-1 ASSET MANAGEMENT FRAMEWORK



2. Objectives

The objective of this manual for Capital Investment Decision Making is to ensure capital and major operating investment decisions deliver new, modified or replacement assets on time, within budget, are fit-for-purpose and capable to meet business objectives.

This objective is accomplished through:

- Optimising lifecycle costing and operational performance.
- New or upgraded assets are fit-for-purpose, cost effective and safe.
- Assets will be reliable and support sustainable operation and maintenance.
- Assets are well supported in the supply chain.
- Ensuring the required capability is delivered by involving stakeholders through a careful assessment of the options, the associated functional requirements and risks.
- Comply to the Glencore Copper Capital Management Framework (CMF) for capital investment decision-making when certain thresholds are met.
- Achieving the best possible Return on Investment (ROI) for stakeholders.
- Articulate the value a project will create or the risk which it will reduce for the business.

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- Ensuring capital projects support the requirements of the Business and Strategic Asset Management Plans.
- Defining, communicating and solving problems as they arise.
- Having the right people in place to enable decision making.
- Having the right information in place to enable decision making.

2.1 Accountabilities

Dedicated resources must be assigned and capabilities developed within each Operation to deliver a Capital Investment Decision Making process to enable effective delivery of the Asset Management objectives.

Assign accountability for each or all the following:

- Ensuring investment in new, modified and replacement assets conform to the CMF.
- Ensure roles are assigned to meet the requirements of the CMF.
- Ensure people with assigned roles have the required authority levels as well as current training and competence to comply with the CMF.

2.2 Scope

This manual is to be read in conjunction with the following:

- Capital Management Framework (CMF)
- Operational Readiness Standard (in development)
- Corporate Risk Framework
- Element E01 Asset Management Leadership
- Element E02 Personnel Management
- Element E03 Asset Management Planning
- Element E05 Equipment Operating and Maintenance Strategy
- Element E06 Commissioning and Handover Management
- Element E13 Procurement and Supply Chain
- Element E14 Risk Management and Compliance
- Element E15 Asset Cost Control
- Element E17 Continuous Improvement
- Element E18 Asset Management Monitoring, Auditing and Assurance.

The scope includes:

- Achieving forecasted business case objectives for new or modified assets.
- Delivering new, modified and fit-for-purpose replacement assets, on time, within budget and ready to operate.
- Ensure investment decisions include and evaluate all required factors such as sustainability.
- Identifying risks such as data quality and accuracy and implementing suitable controls.
- Documenting and communicating to Operations the asset's capability limits, including Standard of Operation.
- Documenting each project's success and making learnings available for future projects.

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3. Definitions

Capital Investment Decision Making – the process for evaluating and analysing scenarios for an organisation's capital investment decisions as to acquire and/or replace assets.

Capital Procurement Item – Items purchased using capital expenditure (CAPEX) that can be depreciated. Generally, equipment and/or components that are routine purchase or "like for like" replacements, e.g. Mobile equipment, Transformer.

Design for Manufacturing and Assembly – a process where assets are designed to be manufactured and assembled easily and economically.

FMEA in Design (DFMEA) – a process for evaluating a design for reliability and robustness against potential failures.

Life Cycle Cost Analysis – an asset's total cost throughout its life, including the costs of planning, design, acquisition, operations, maintenance, and disposal, less any residual value.

Material Risk – a designation (typically in a particular regulatory context) indicating a certain risk is of sufficient significance to an organization and must be managed following certain minimum criteria.

Project Execution – the process to Safely deliver the project on time and on budget.

Project Feasibility – the process of ensuring the project is ready for execution by detailing the selected option to minizine execution risk.

Project Pre-feasibility – the process of conducting option trade off studies, project competitiveness evaluation, single option selection, initial design and the development of a well-defined investment case.

Project Concept – the process of identifying options, analysis of option viability, reducing option and developing investment assessment.

Intermediate Project – Project with a:

- Total cost between US$20M and US$150M **or**,
- Total cost between US$1M and US$20M with a complexity score of >13

These are typically either a mid-range CAPEX Greenfield Project or a complex low range CAPEX Project, e.g. Acid Plant, Surface Crushing Facility, Major Plant Turnaround (PTA), etc.

Major Project – Project with a total cost >US$150M. These are typically large Greenfield or Brownfield Projects, e.g. new Mining, Milling or Smelter complex, etc.

Minor Project – Either a Project with a:

- Total cost between US$1M and US$20M and a complexity score < 13 **or**,
- Total cost <US$1M.

These are typically relatively simple low range CAPEX Projects, e.g., Primary Ventilation Fan, Thickener Upgrade, etc.

RAMS – Reliability, Availability, Maintainability and Safety are system design attributes with substantial potential impacts on the lifecycle cost and performance of an engineered system. In RAM modelling, Reliability is the probability an item will survive after a determined time. Availability is the percentage of time a system is considered ready to use when tasked. Maintainability is a function of the failure identification time, logistics capabilities, diagnostic capabilities, tooling and instruments, repair capabilities and Safety to eliminate or minimise accidents and enhance safety of personnel and the asset itself.

Tollgate – a critical hold point to ensure that the project is progressing as planned.

Value Management – a structured and analytical process in which a prescribed works plan is followed to achieve best value and, where appropriate, best value for money in products, processes, services, systems and organisations. The process may be applied to management decision making at any level of an organisation.

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4. Approach

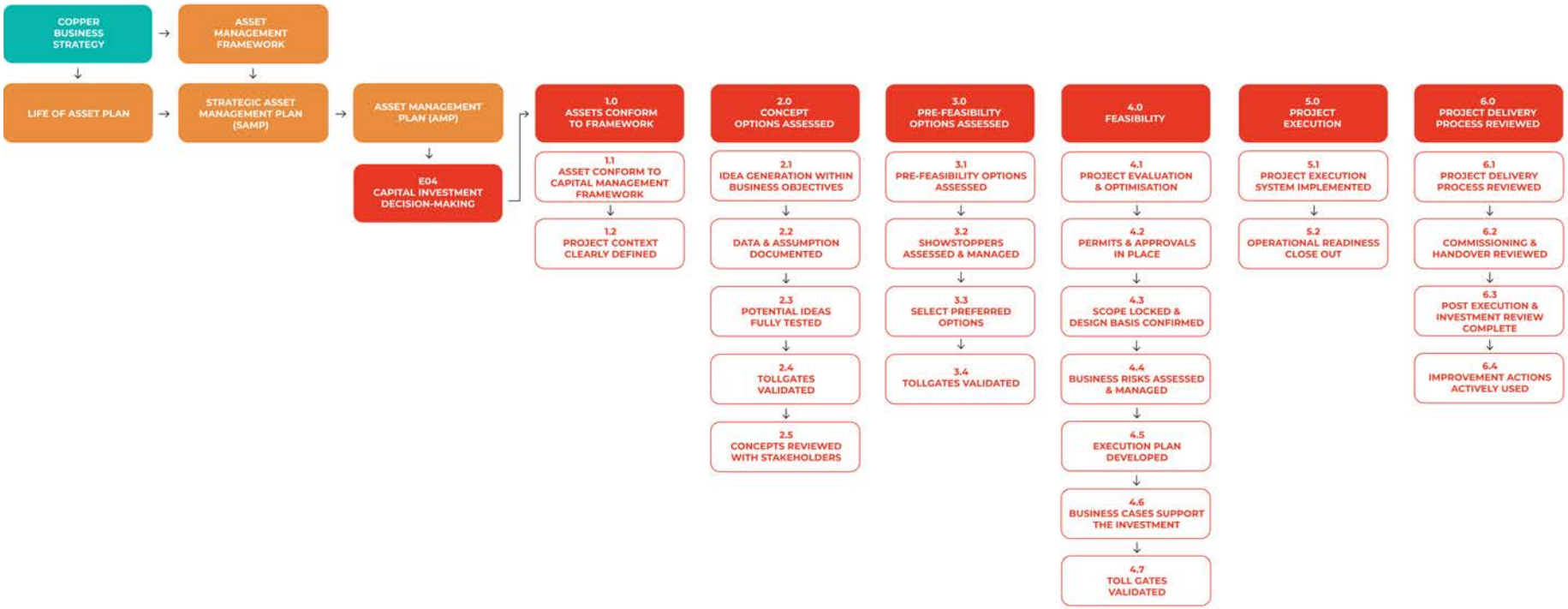
The approach for Capital Investment Decision Making is described through:

- Process Flow – pictorially illustrates the key Capital Investment Decision Making activities, refer to Figure 4-1.
- Process Description – defines in more detail the key Capital Investment Decision Making activities, refer to Section 4.2.
- Process RACI Matrix – indicates for each activity the Functions, who are Responsible, who are Accountable, who are Consulted prior to final decision/action and those Informed of any outcomes and actions, refer to Section 4.3.

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4.1 Process Flow

FIGURE 4-1 CAPITAL INVESTMENT DECISION MAKING PROCESS



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4.2 Process Description

TABLE 4-1 CAPITAL INVESTMENT DECISION MAKING PROCESS DESCRIPTION

PROCESS STEP	DESCRIPTION
1.0 Assets Conform to the Framework	**Deliver new, modified or replacement assets which conform to Glencore Copper's Capital Management Framework (CMF), purchasing and supply requirements as well as specific asset management key requirements.**
1.1 Asset Conform to the Capital Management Framework	Asset acquisitions and major modifications need to conform to the requirements within the Glencore Copper Capital Management Framework. The Glencore Copper Operational Readiness Guideline is also to be consulted to ensure effective and efficient project implementation and rapid start up post project commissioning. The Strategic Asset Management Plan should also be consulted to ensure the project contributes to the asset management objectives. Element E13 Procurement and Supply Chain Management should be consulted regarding purchasing and supply chain requirements.
1.2 Project Context Clearly Defined	The context of the project to be clearly defined, context could include: • Geographical Location – identify potential project execution constraints and risk sources such as: – Local weather/climate extremes – Geo-technical and topographical issues – Site access constraints – Utilities and local service availability – Environmental sensitivities – Human and material resource availability. • Regulatory and Legal Requirements – may impact project execution performance by placing conditions on certain parameters such as: – Prioritisation of standards – Local content requirements – Adherence to site specific and local environmental regulations – Adherence to Glencore Copper Code of Conduct – Understanding restrictions on human and resource availability. • Cultural Environment – dedicates how the project management team may need to adjust the project execution plan, by considering: – Personnel accommodation and work facilities – Local holidays and acceptable working hours – Restricted or protected areas – Security considerations and requirements. • Technology, skills and experience for the project – Availability of technological know-how can affect the both the objectives and values of a project through its impact on: – Design complexity – Human resource availability – Speed and efficiency in project execution – Adherence to scope and standards – Reliability and operability of the end-product – Safety in project execution and operation.

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PROCESS STEP	DESCRIPTION
	• What are the market conditions in which the project exists? – Market conditions are always dynamic and, depending on the overriding economic environment, can be either beneficial or detrimental to the performance of a project. Varying market conditions may affect a project in the following ways: – Ability to finance the project – Human and material resource availability – Changes to project scope and/or standards. • Who are the project stakeholders, what influence do they have and who is controlling the project? – A project is inevitably affected by the influences exerted on it by its controlling organisation and other stakeholders. The extent of this influence is generally determined by the following stakeholder factors: – Experience – Culture – Style – Structure – Maturity – Risk Attitude – Interests and Priorities.
2.0 Concept Options Assessed	**Progress ideas or opportunities in the concept phase with the intent to achieve the strategic plan and objectives.**
2.1 Idea Generation Supporting Business Objectives	Business objectives are laid out in the Life of Asset and Strategic Asset Management Plans. These plans detail both business and asset management objectives and capital investment options are to be derived from these objectives. Concept ideas and opportunities are to be identified for consideration to enable delivery of the business and strategic objectives. All ideas and opportunities identified to be documented.
2.2 Data and Assumptions Documented	Concept data to be acquired and assumptions documented for the next stage. Depending on value, complexity and risks, documentation may include Concept Study Report, Scope of Work and Project Risk Assessment.
2.3 Potential Ideas Fully Tested and alternative options compared	Opportunities identified in the concept stage shall be tested to ensure their contribution to relevant business and asset management objectives. Consult with key stakeholder to conduct high level analysis and ranking of each opportunity considering at least the following: • Legislative compliance or licence to operate factors • Risk reduction based on technical assessment • Financial benefits based on financial modelling, including revenue opportunity and whole of life cost estimate. Cost estimate and business case requirements to be in line with the CMF requirements and any relevant benchmarking. • Develop a definition and scoring matrix for all the relevant objectives and requirements.
2.4 Tollgates Validated	Tollgate to be conducted to determine: • Review objectives of the tollgate • Changes to the project charter • Project benefit update • Review project status

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PROCESS STEP	DESCRIPTION
	• Tollgate decision • Action items and next steps. Tollgates ensure: • Projects are kept on track • Stop non effective projects • Facilitates better results.
2.5 Concepts Reviewed with Stakeholders	A plan for managing and communicating with stakeholders is to be developed and all concepts identified shall be reviewed with the stakeholders. • Develop key stakeholder and RACI matrix • Develop communication plan • Record stakeholder authorisation.
3.0 Pre-feasibility Options Assessed	**Generating and reviewing pre-feasibility options to achieve business requirements and selecting the preferred option for further analysis.**
3.1 Pre-feasibility Options Assessed and Operational Readiness Initiated	Items selected in the concept stage are to be assessed and compared to alternative options. The assessment is to cover as far as possible the predicted cost and risk associated with the whole of life of the asset. The Operational Readiness Plan should commence being developed in Pre-feasibility for Minor, Intermediate and Major Projects. The Operational Readiness Plan ensures that Operational Readiness considerations are included in the project scope and are appropriately planned and funded to be undertaken in Execution. The plan also helps ensure business continuity after the project is completed, it secures value preservation by ensuring the proper allocation of resources and a fit-for-purpose approach to the operation of a new piece of equipment or new project.
3.2 Showstoppers Assessed and Managed	Defining the project context provides an opportunity in the pre-feasibility stage to identify risks that are possibly preventing the project from transitioning to the feasibility stage. Risks potentially impacting on business and asset management objectives are to be identified. Key project performance indicators are to be established to ensure risk mitigation and best project option selection.
3.3 Select Preferred Option	Select the preferred option to proceed to the 'feasibility' stage.
3.4 Tollgates Validated	Tollgate to be conducted to determine: • Review objectives of the tollgate • Changes to the project charter • Project benefit update • Review project status • Tollgate decision • Action items and next steps.
4.0 Feasibility	**Optimise the preferred option to proceed to the 'execution' stage**.
4.1 Project Evaluation and Optimisation	Rigorous evaluation and optimisation of the project to increase accuracy of projected risk, cost and performance. The project's return on investment must meet the Glencore Copper threshold or specified requirement.

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PROCESS STEP	DESCRIPTION
4.2 Permits and Approvals in Place	All permit, licence and approval applications are to be developed and lodged as required in enough time to understand the material risk.
4.3 Scope Locked and Design Basis Confirmed	The project scope is to be finalised, approved, locked and design basis confirmed such as: • Total cost of ownership (Life Cycle Cost Analysis) • Design Failure Mode and Effects Analysis (FMEA) to mitigate failures • Design for Operability and Constructability • Design for Reliability, Availability, Maintainability, Safety and Sustainability (RAMS^2) • Value Management (VM).
4.4 Business Risks Assessed and Managed	Risks are to be identified, evaluated and managed through the Glencore Copper Risk Management Framework, CU-RISK-GBL-PRO-003 Project Risk Management Procedure and the Capital Management Framework (CMF). Asset Budget Attainability Guideline should also be consulted.
4.5 Execution Plan Developed	A Project Execution Plan is to be developed to cover: • The scope of all feasibility activities proposed. • The objectives and benefits of completing these activities prior to project approval with consideration to the issue of continuity of key people. • The deliverables, which will result from the work plan. • A detailed budget and schedule to execute the proposed work plan. • Roles and responsibilities of key team members. • Risks associated with the works, in addition to mitigation strategies for key risks. • Procedures and systems to be employed including reporting requirements. • Finalise the Operational Readiness plan.
4.6 Business Cases Support the Investment	The business case is to be validated so value is optimised while meeting business needs within the acceptable level of risk.
4.7 Toll Gates Validated	Tollgate to be conducted to determine items such as: • Review objectives of the tollgate • Changes to the project charter • Project benefit update • Review project status • Project execution plan reviewed and signed off • Business Case final sign-off • Tollgate decision • Action items and next steps. Refer to Glencore Copper CMF regarding stage gate process.
5.0 Project Execution	**New or modified asset deliverables prior to handover to operations.**
5.1 Project Execution System Implemented	The following activities are to be conducted as part of the project execution in line with the CMF and site requirements: • Develop and align a Project Performance System for safe, on time and on-budget execution. • Implement an effective Project Management Operating System aligning all project execution teams on key process indicators. • Prepare reporting to demonstrate progress, identify risks and trends.

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PROCESS STEP	DESCRIPTION
5.2 Operation Readiness Close Out	Refer to the Operational Readiness Standard. The final stage of Operational Readiness is to be conducted and closed out.
6.0 Project Delivery Process Reviewed	**Reviewing new or modified asset performance and the project delivery process.**
6.1 Project Delivery Process Reviewed	On completion of project handover, a formal post project process compliance and quality review is to be carried out. If deficiencies are identified, then consideration for project delivery improvement should be given.
6.2 Commissioning and Handover Reviewed	Review the commissioning and handover activities and refer to Element E06 Commissioning and Handover Management to review against process.
6.3 Post Execution and Investment Review Completed	A post investment review is to be carried out to ascertain project investment performance against the business case and budget. A review to be conducted post project execution to determine project execution success. All data is to be collected during the review and lesson learnt captured.
6.4 Improvement Actions Actively Used	Processes for project execution and business performance improvement actions are to be established and used.

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4.3 Process RACI Description

The RACI matrix below in Table 4-2 addresses the high-level functional elements of the Capital Investment Decision Making process. For the overall Asset Management RACI matrix, refer to the Asset Management Intranet page.

Note the HSE and Commercial Departments do not appear in this high-level RACI, although their involvement will be required from time to time dependent on the nature of the variance.

The codes used in the RACI analysis are defined as follows:

TABLE 4-2 PROCESS RACI

R	**Responsible**	**The "doer" – the individual(s) who will perform an activity or directs others to perform the activity.**
A	**Accountable**	"The buck stops here" –the individual who ultimately "owns" the activity – there should only be one person.
C	**Consulted**	A person who needs to provide feedback and contribute to the activity.
I	**Inform**	The individual(s) who must be informed after a decision or action is taken if relevant. (via one-way communication).

Function / Process Step	General Management	Maintenance Management	Operations Management	Maintenance Execution	Supply / Warehouse	Maintenance Planning	Technical Services	Projects / Engineering
1.0 Assets Conform to Framework								
1.1 Asset Conform to Capital Management Framework	C	A	C	C	C	C	C	R
1.2 Project Context Clearly Defined		A	C	R	C		R	R
2.0 Concept Options Assessed								
2.1 Idea Generation Within Business Objectives		A	C	R	C	C	R	R
2.2 Data and Assumptions Documented								
2.3 Potential Ideas Fully Tested and alternative options compared		A	C	R	C	C	R	R
2.4 Data and Assumptions Documented		A	C	R	C	C	R	R
2.5 Tollgates Validated		A	C	R	C		R	R
2.6 Concepts Reviewed with Stakeholders	C	A	C	R			R	R
3.0 Pre-feasibility Options Assessed								
3.1 Pre-feasibility Options Assessed		A	C	R	C		R	R
3.2 Showstoppers Assessed and Managed	C	A	C	R	C		R	R
3.3 Select Preferred Option								

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Process Step \ Function	General Management	Maintenance Management	Operations Management	Maintenance Execution	Supply / Warehouse	Maintenance Planning	Technical Services	Projects / Engineering
3.4 Tollgates Validated		A	C	R	C		R	R
4.0 Feasibility								
4.1 Project Evaluation and Optimisation		A	C	R	C		R	R
4.2 Permits and Approvals in Place	C	A	C	R	C		R	R
4.3 Scope Locked and Design Basis Confirmed		A	C	R	C		R	R
4.4 Business Risks Assessed and Managed	A	R	R	R	R	C	R	R
4.5 Execution Plan Developed		A	C	R	C		R	R
4.6 Business Cases Support the Investment	C	A	C	R	C		R	R
4.7 Toll Gates Validated		A	C	R	C		R	R
5.0 Project Execution								
5.1 Project Execution System Implemented	C	A	R	R	R	R	R	R
5.2 Operation Readiness Close Out		A	R	R	R	R	R	R
6.0 Project Delivery Process Reviewed								
6.1 Project Delivery Process Reviewed		A	R	R	C	C	R	R
6.2 Commissioning and Handover Reviewed		A	R	R	C	C	R	R
6.3 Post Execution and Investment Review Completed	C	A	R	R	C	C	R	R
6.4 Improvement Actions Actively Used	C	A	R	R	C	C	R	R

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TABLE 4-3 FUNCTIONAL DESCRIPTIONS

Function	Description
General Management	• Ensures business risks are assessed and managed throughout the Capital Investment Decision Making process. • Provides input with regards to new, modified or replacement assets conform to the CMF. • Provides input with regards to the review of concepts. • Provides input with regards to the development of business cases to support the investment. • Provides input with respect to the implementation of a project execution system. • Provides input with respect to post execution and investment reviews. • Provides input with respect to improvement actions.
Maintenance Management	• Ensures the full spectrum of the Capital Investment Decision Making process is implemented, management and improved. • Responsible with regard to business risks which are assessed and managed throughout the Capital Investment Decision Making process.
Maintenance Execution	• Responsible for the implementation, management and improvement of the full spectrum of the Capital Investment Decision Making process. • Provides input with respect to risks are assessed and managed throughout the Capital Investment Decision Making process.
Maintenance Planning	• Provides input with respect to risks are assessed and managed throughout the Capital Investment Decision Making process. • Provides input with respect to the generation of ideas. • Provides input with respect to the testing of ideas and comparing with alternate options. • Provides input with respect to documenting data and assumptions. • Provides input with respect to business risks are assessed and managed throughout the Capital Investment Decision Making process. • Responsible for project execution delivery and equipment capability management. • Provided input with respect to project delivery process reviews.
Operations Management	• Provides input with respect to risks are assessed and managed throughout the Capital Investment Decision Making process. • Provides input with respect to the generation of ideas. • Provides input with respect to the testing of ideas and comparing with alternate options • Provides input with respect to documenting data and assumptions. • Provides input with respect to the review of concepts. • Provides input with respect to assessing pre-feasibility options and managing showstoppers. • Provides input with respect to validating tailgates. • Provides input with respect to the feasibility phase. • Responsible with regards to business risks are assessed and managed. • Responsible with regards to project execution. • Responsible with regards to the review of project delivery processes.

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Function	Description
Supply/Warehouse	• Provides input with respect to assets conforming to the Capital Management Framework • Provides input with respect to the generation of ideas. • Provides input with respect to the testing of ideas and comparing with alternate options • Provides input with respect to documenting data and assumptions. • Provides input with respect to validating tailgates. • Provides input with respect to assessing pre-feasibility options and showstoppers. • Provides input with respect to the feasibility phase. • Responsible with respect to business risks are assessed and managed. • Responsible with regards to project execution. • Responsible for project execution delivery and equipment capability management.
Projects Engineering	• Responsible for the implementation, management and improvement of the full spectrum of the Capital Investment Decision Making process. • Provides input with respect to risks are assessed and managed throughout the Capital Investment Decision Making process.
Technical Services – Mining, Processing	• Responsible for the implementation, management and improvement of the full spectrum of the Capital Investment Decision Making process. • Provides input with respect to risks are assessed and managed throughout the Capital Investment Decision Making process.

5. Deployment

The deployment of the Capital Investment Decision Making approach across all business areas is through:

- Awareness presentation
- Position description alignment
- Competency assessment
- Training, coaching and mentoring.

5.1 Awareness Presentation

This Manual is supported by a slide presentation and video to help facilitate communication and to raise awareness on how it contributes to the Asset Management Framework and business success.

5.2 Position Description Alignment

To effectively deploy this Manual, the essential process steps and activities need to be diligently mapped and assigned to approved positions within Glencore Copper.

The RACI matrix assists with the visualisation, so everyone is working towards the same goal. It clarifies, an individual's responsibilities, and also those of others. It has proven to be an effective tool to quickly understand and close the gaps between current and desired practices and business process activities.

Position descriptions including KPIs need to be aligned to the RACI matrix and measures of success. This is the basis for ongoing performance reviews and the creation and execution of individual development plans.

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5.3 Competency Assessment

This Manual is supported by an Asset Management competencies assessment tool which is used to:

- Guide recruitment, selection, development, and deployment of people in asset management roles across the company.
- Provide employees responsible for managing assets with feedback, development, coaching and recognition for their performance.
- Map, build and manage the asset management talent pool across the company.
- Identify strengths and development needs in asset management practitioners.

5.4 Training, Coaching and Mentoring

Coaching and mentoring are another way to assist employees improve their understanding of their role and to improve performance.

Coaching is generally about improving short-term performance where mentoring is about achieving high performance and career aspirations over a longer term.

Coaching or mentoring requires addressing where a person is not currently up to the required level to perform their role as effectively and efficiently as they could. The build-up of skills to perform efficiently and effectively requires a large investment of time and cooperation.

Coaches and Mentors need to be carefully selected and must be committed for the long term to effect maximum success.

6. Measures of Success

Refer to Glencore Copper's AM Manual E16 Reporting and Performance Management Manual for current KPIs.

These are typical project metrics; each project will have its own metrics based on the project requirements. Refer to the CMF for performance measure dashboard, which should also be considered.

TABLE 6-1 MEASURE OF SUCCESS

Metric	Return on Investment (ROI)
Definition	A financial ratio used to calculate the benefit an investor will receive in relation to their investment cost.
Performance Indicator	As per operational requirements.
Calculation	ROI = ((Gain from Investment – Cost of Investment) / (Cost of Investment)) x 100.
Rationale	Return on Investment (ROI) is a performance measure used to evaluate the returns of an investment or to compare the relative efficiency of different investments.
Action	Project not to advance should the required ROI threshold not be achievable.

Metric	Internal Rate of Return (IRR)
Definition	IRR is a financial analysis method used to estimate the profitability of potential investments.
Performance Indicator	As per operational requirements.
Calculation	$IRR = R1 + \frac{(NPV1\ x\ (R2-R1)}{(NPV1-NPV2)}$

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	R1, R2 = randomly selected discount rates NPV1 = higher net present value NPV2 = lower net present value
Rationale	IRR is a discount rate making the net present value (NPV) of all cash flows equal to zero in a discounted cash flow analysis.
Action	Project not to advance should the required IRR threshold not be achievable.

Metric	**Payback Period**
Definition	The length of time needed to recover the funds expended, or to break-even.
Performance Indicator	As per operational requirements.
Calculation	Payback Period $= \frac{Initial\ Investment\ (\$)}{Net\ Annual\ Cash\ Flow\ (\$)}$
Rationale	The shorter the payback period the more profit will be realised from a project.
Action	Project not to advance should the required payback period threshold not be achievable.

Metric	**Project Budget Compliance**
Definition	Actual Budget vs Planned Budget.
Performance	As per operational requirements.
Calculation	(Actual Budget/Planned Budget)/100.
Rationale	This is important as it is used to monitor the effectiveness of the shutdown planning and scheduling process and the compliance to the planned budget for the shutdown.
Action	Project budget to be tracked and actions implemented to bring budget back to planned should it deviate from planned.

Metric	**Project Schedule Compliance**
Definition	% of scheduled tasks compared to the total project tasks completed.
Performance Indicator	100% Compliance to Schedule.
Calculation	[Number of scheduled tasks completed in the project / Total number of tasks planned in the project] x 100.
Rationale	This is important as it is used to monitor the effectiveness of the project planning and scheduling process and the compliance to the planned budget for the project.
Action	Project schedule to be tracked and actions implemented to bring budget back to planned should it deviate from planned.

7. Improvement

The process to review and continually improve the approach and deployment for Capital Investment Decision Making is scheduled Annually, or when there is a sudden change in the Operating Context. As the organization matures a longer review period could be considered.

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8. References

The following documents provide detailed support material and should be read in conjunction with this Manual to provide a comprehensive understanding of Capital Investment Decision Making.

TABLE 8-1 REFERENCES

Document Title	Description
Asset Management Framework	This document defines the overall direction of asset management across Glencore Copper.
Glencore Copper Capital Management Framework	Details the Capital Management requirements for Glencore Copper laid out in three parts for minor, intermediate and major capital projects.
Institute of Asset Management SSG 6&7 Capital Investment, Operations and Maintenance Decision Making	Provides guidance regarding Capital Investment Decision Making.

9. Revision History

Version No.	Reason for Revision	Date
0.1	New Manual to support the Asset Management Framework	15/11/2021
0.2	Reformat and Issue for Final Review	19/05/2022
1.0	Update Site Signoff and Publish	30/05/2022

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10. Appendix A – Stakeholder Signoff

Stakeholder Name	Site or Department	Contact	Stakeholder Signoff
Anthony Green	Asset Management - Australia	Anthony.green@glencore.com.au	7/6/2022
Akeem Araoye	Asset Management – North America	Akeem.Araoye@glencore.ca	7/6/2022
Yuri Yurac	Asset Management – South America	Yuri.yurac@glencore.cl	8/6/2022
Lawrence Hlatshwayo	Asset Management – Africa	Lawrence.hlatshwayo@glencore.co.za	10/6/2022
Rohan Reisener	Asset Management - Australia	Rohan.reisener@glencore.com.au	7/6/2022
Dino Moll	Asset Management - Australia	Dino.Moll@glencore.com.au	13/6/2022

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Notes

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Capital Investment Decision-Making

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Capital Investment Decision-Making

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